

SLEETMUTE, AK, USA

WMO: 703407

Lat: 61.717N

Lon: 157.150W

Elev: 54

StdP: 100.67

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26553

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.6	-34.9	-34.0	0.2	-29.6	-33.7	0.2	-29.1	6.9	-3.6	5.7	-9.1	0.2	10	0.679

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	23.7	14.8	22.1	14.1	20.0	13.3	16.4	21.6	15.2	19.6	14.2	18.4	1.5	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	10.1	17.8	12.9	9.3	16.7	12.3	8.9	16.0	45.7	21.2	42.3	19.7	39.7	18.6	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
5.1	4.5	3.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			-33.6	17.5	1.9	1.6	-35.0	18.7	-36.1	19.6	-37.1	20.5	-38.5	21.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.2	-18.5	-12.3	-9.1	0.3	7.5	12.6	13.9	12.2	7.7	-1.5	-10.5	-17.3
	DBStd	13.08	10.29	8.57	7.19	4.82	3.90	2.82	2.67	2.55	3.11	5.78	7.11	8.72
	HDD10.0	4399	884	626	591	290	96	7	2	9	78	356	616	846
	HDD18.3	7131	1143	859	849	540	335	173	141	189	318	614	866	1104
	CDD10.0	314	0	0	0	0	19	85	122	77	10	1	0	0
	CDD18.3	4	0	0	0	0	0	1	3	0	0	0	0	0
	CDH23.3	88	0	0	0	0	4	31	52	1	0	0	0	0
	CDH26.7	18	0	0	0	0	0	6	12	0	0	0	0	0
Wind	WSAvg	1.1	0.9	1.0	1.4	1.4	1.6	1.2	1.1	1.0	0.9	1.1	1.0	0.9
Precipitation	PrecAvg	693	51	44	39	38	40	55	78	89	80	63	59	58
	PrecMax	847	111	91	84	61	77	80	116	135	138	123	129	95
	PrecMin	541	14	5	8	10	11	29	37	39	23	13	5	21
	PrecStd	80	22	19	19	12	14	14	18	24	30	25	31	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.8	5.6	7.2	14.1	23.5	27.0	28.0	22.7	18.9	11.7	6.0	2.9
		MCWB	1.5	1.8	2.4	6.1	12.4	16.1	18.0	16.3	12.3	8.6	2.1	1.5
	2%	DB	1.3	2.9	4.4	12.3	21.2	23.4	24.1	20.3	16.0	8.8	2.5	0.9
		MCWB	0.1	0.4	0.9	5.3	11.5	14.1	15.9	14.5	10.5	6.9	0.8	-0.2
	5%	DB	-0.3	1.3	2.7	10.2	18.0	21.9	22.2	18.2	14.0	7.5	1.1	-1.5
		MCWB	-0.9	-0.2	0.2	4.4	10.0	13.2	15.2	13.5	10.1	5.6	-0.2	-2.0
	10%	DB	-2.8	-0.9	1.3	8.0	15.2	19.2	19.9	17.1	12.6	6.2	-1.0	-4.2
		MCWB	-4.0	-1.9	-0.5	3.3	8.0	12.4	14.3	12.8	9.6	4.5	-1.5	-5.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	1.9	3.3	6.9	13.4	17.2	19.1	17.3	13.0	9.1	2.6	1.9
		MCDB	3.8	4.7	5.3	12.6	21.6	24.8	26.2	20.4	16.5	11.2	5.2	3.1
	2%	WB	0.3	0.8	1.7	5.6	11.8	15.2	16.9	15.5	11.7	7.6	0.9	-0.2
		MCDB	1.4	2.6	3.6	11.2	19.7	21.3	22.8	18.6	14.1	8.6	2.6	0.6
	5%	WB	-0.9	0.0	0.6	4.6	10.3	14.0	15.7	14.4	10.8	6.1	-0.1	-2.0
		MCDB	-0.1	1.5	2.6	9.6	17.1	19.9	20.6	17.5	13.3	7.3	0.9	-1.7
	10%	WB	-3.4	-1.9	-0.5	3.6	8.9	13.0	14.7	13.4	10.0	4.7	-1.5	-4.8
		MCDB	-2.4	-0.7	1.3	7.9	14.6	18.7	19.0	16.2	12.5	6.2	-0.9	-3.8
Mean Daily Temperature Range	5% DB	MDBR	8.0	9.8	10.5	10.3	10.9	10.5	8.8	8.0	8.0	6.3	6.8	8.0
		MCDBR	8.3	8.7	9.8	14.0	15.6	14.9	14.3	11.2	10.2	7.0	6.2	8.7
	5% WB	MCWBR	7.3	7.4	7.3	8.5	8.1	7.1	7.2	6.6	6.5	5.1	5.4	8.2
		MCWBR	8.3	8.5	8.2	12.7	14.3	12.9	12.1	9.3	8.6	6.0	6.1	8.6
Clear-Sky Solar Irradiance	taub	taub	0.220	0.268	0.275	0.343	0.366	0.376	0.376	0.381	0.318	0.292	0.228	N/A
		taud	2.072	2.229	2.273	2.165	2.331	2.351	2.392	2.381	2.543	2.430	2.081	N/A
	Ebn at Noon	Ebn at Noon	493	710	850	839	853	848	839	799	805	687	462	N/A
		Edh at Noon	36	63	86	119	110	110	104	97	68	53	34	N/A
All-Sky Solar Radiation	RadAvg	0.31	0.98	2.83	4.52	5.12	5.25	4.40	3.43	2.31	1.05	0.39	0.15	
	RadStd	0.05	0.14	0.35	0.41	0.57	0.40	0.41	0.39	0.33	0.12	0.06	0.02	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	+1.00	N/A	+0.85	N/A	N/A	N/A	N/A	-299	-358	N/A	N/A

Nomenclature: See separate page